

PROJECT DOCUMENTATION

Project Title

To-Do List Application

Internship

DecodeLabs Python Internship 2026

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Project Overview

The To-Do List Application is a simple command-line program developed using Python. It allows users to add tasks, view stored tasks, and exit the application through a menu-driven interface. The project demonstrates the practical implementation of Python Lists, Dictionaries, Functions, and Enumerate.

Problem Statement

Develop a Python program where users can:

- Add tasks
- View tasks
- Exit the application

The project should utilize Lists, Dictionaries, Append Operations, Functions, and Enumerate.

Objective

The main objective of this project is to understand how multiple items can be stored and managed using Python data structures while creating a basic task management system.

Concepts Used

List

A list is used to store all tasks.

Example:

```
tasks = []
```

Dictionary

Each task is stored as a dictionary.

Example:

```
{"task": "Finish Python Assignment"}
```

Functions

Functions are used to organize the code into reusable blocks.

Functions used:

- `add_task()`
- `view_tasks()`
- `main()`

Enumerate

The `enumerate()` function is used to display task numbers while viewing tasks.

Program Workflow

1. The program displays a menu.
2. The user selects an option.
3. If the user chooses Add Task:
 - A task name is entered.
 - A dictionary is created.
 - The dictionary is appended to the list.
4. If the user chooses View Tasks:
 - All tasks are displayed using `enumerate()`.
5. If the user chooses Exit:
 - The program terminates.
6. Invalid inputs are handled with an error message.

Functions Description

`add_task(task_name)`

Purpose:

Adds a new task to the task list.

Process:

- Creates a dictionary containing the task.
- Appends the dictionary to the tasks list.
- Displays a success message.

view_tasks()

Purpose:

Displays all stored tasks.

Process:

- Checks whether the task list is empty.
- Uses enumerate() to display task numbers.
- Prints all available tasks.

main()

Purpose:

Controls the overall program flow.

Process:

- Displays menu options.
- Accepts user input.
- Calls the required functions.
- Continues execution until the user exits.

Sample Execution

To-Do List Application

1. Add Task
2. View Tasks
3. Exit

Enter your choice (1/2/3): 1

Enter the task name: Finish Python Assignment

Task 'Finish Python Assignment' added successfully.

Enter your choice (1/2/3): 2

Tasks:

1. {'task': 'Finish Python Assignment'}

Testing

Test Case 1

Input:

Add Task

Task:

Finish Python Assignment

Expected Output:

Task added successfully.

Result:

Passed

Test Case 2

Input:

View Tasks

Expected Output:

Displays all saved tasks with numbering.

Result:

Passed

Test Case 3

Input:

Exit

Expected Output:

Program exits successfully.

Result:

Passed

Test Case 4

Input:

Invalid Menu Choice

Expected Output:

Invalid choice. Please try again.

Result:

Passed

Advantages

- Simple and easy to understand.
- Demonstrates core Python concepts.
- Uses structured data storage.
- Menu-driven interface improves usability.
- Beginner-friendly project.

Conclusion

The To-Do List Application was successfully developed using Python. The project effectively demonstrates the use of Lists, Dictionaries, Functions, Append Operations, and Enumerate. All project requirements were successfully implemented, and the application performs task management operations as expected.